

Gradient-Composite Flatness: Why Affine Stationarity Is Exact but Does Not Close the Canonical UBT Branch

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Abstract

Consider the auxiliary gradient-composite restriction in which a Lorentz-real field is written as $\Theta = Y^a \mathbf{u}_a$ and the tetrad is defined by $e^a = \mathcal{N}_0^{-1/2} dY^a$. We prove the stronger statement hidden behind the previously reported affine-stationarity calculation: every non-degenerate configuration of this restriction is locally a pullback of Minkowski space and therefore has identically vanishing Riemann curvature. The Hilbert–Palatini term vanishes identically, while the locked kinetic and cosmological terms are multiples of the Jacobian volume and are null Lagrangians under compactly supported variations. Hence the affine representer is stationary for all coefficients, but only as a corollary of a flatness no-go. This auxiliary ∂ -gradient restriction cannot be called the surviving GR branch of canonical UBT, whose tetrad is $E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta$ and whose self-consistent D -composite variation remains open.

1 Auxiliary gradient-composite restriction

On the Lorentz slice write

$$\Theta = Y^a \mathbf{u}_a, \quad e^a{}_\mu = \mathcal{N}_0^{-1/2} \partial_\mu Y^a, \quad e^a = \mathcal{N}_0^{-1/2} dY^a, \quad (1)$$

with $\det(\partial_\mu Y^a) \neq 0$. The induced metric is

$$g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu = \mathcal{N}_0^{-1} \eta_{ab} \partial_\mu Y^a \partial_\nu Y^b. \quad (2)$$

This is an auxiliary restriction obtained by replacing the canonical covariant jet $D_\mu \Theta$ by the exact gradient $\partial_\mu \Theta$.

Theorem 1 (Exact-gradient flatness no-go). *Every nondegenerate metric of the form (2) is locally flat:*

$$R^\rho{}_{\sigma\mu\nu}[g] = 0, \quad R^{ab}[\tilde{\omega}(e)] = 0. \quad (3)$$

Consequently the gradient-composite restriction cannot represent a generic curved GR geometry.

Proof. Nondegeneracy makes $Y = (Y^0, Y^1, Y^2, Y^3)$ a local coordinate map. In the coordinates Y^a the metric is simply $\mathcal{N}_0^{-1} \eta_{ab}$, so its Riemann tensor vanishes. Equivalently, the coframe is exact, $de^a = \mathcal{N}_0^{-1/2} d^2 Y^a = 0$. The zero spin connection is metric-compatible and satisfies the torsion-free Cartan equation $de^a + \tilde{\omega}^a{}_b \wedge e^b = 0$; uniqueness of the Levi–Civita spin connection therefore gives $\tilde{\omega}^a{}_b = 0$ in this coframe and $R^{ab} = 0$. $\square$

Corollary 1 (Hilbert–Palatini degeneracy). *For every admissible configuration of the gradient-composite restriction,*

$$\epsilon_{abcd} e^a \wedge e^b \wedge R^{cd}(\tilde{\omega}(e)) = 0. \quad (4)$$

Thus the Einstein/Hilbert–Palatini term carries no local dynamics on this restriction, not merely at the affine background.

2 The volume terms and affine stationarity

The central metric lock gives

$$\langle \partial_\mu \Theta, \partial_\nu \Theta \rangle_\# = \mathcal{N}_0 g_{\mu\nu}, \quad (5)$$

so in four dimensions

$$\frac{1}{2} \sqrt{-g} g^{\mu\nu} \langle \partial_\mu \Theta, \partial_\nu \Theta \rangle_\# = 2 \mathcal{N}_0 \sqrt{-g}. \quad (6)$$

On an orientation-preserving patch,

$$\sqrt{-g} = \det(e^a{}_\mu) = \mathcal{N}_0^{-2} \det(\partial_\mu Y^a). \quad (7)$$

Therefore the complete restricted action with $V \equiv 0$ is a coefficient multiple of the Jacobian integral

$$\int d^4x \det(\partial_\mu Y^a), \quad (8)$$

which is a null Lagrangian: its first variation is a boundary term by the Piola identity. In particular the affine map

$$\Theta_{\text{aff}} = \Theta_0 + \sqrt{\mathcal{N}_0} \bar{E}_\mu x^\mu \quad (9)$$

is stationary under compactly supported Lorentz-real variations for all Λ , κ , and $\mathcal{N}_0$.

Corollary 2 (Correct status of affine admissibility). *The earlier statement called **GAP-10T-COMPOSITE-FLAT** is true only as an auxiliary corollary of Theorem 1. It does not select a physical branch and does not evade the curved-space integrability problem.*

3 Relation to canonical UBT

Canonical UBT defines

$$E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta, \quad (10)$$

where the connection is reconstructed from the tetrad and torsion and hence enters the defining jet self-consistently. Replacing D_μ by ∂_μ removes precisely the non-holonomic structure capable of carrying curvature. Accordingly:

- **GAP-10T-GRADIENT-FLATNESS** is **CLOSED AS NO-GO [L1]**;
- gradient-composite affine stationarity is **CLOSED AS AN AUXILIARY COROLLARY [L1]**;
- the actual self-consistent D -composite variation, curved dynamics, torsion selection, and coefficient origin remain open under **GAP-10T-DYN** and **GAP-10D**.

Scope and non-claims

This note proves a no-go for the exact-gradient restriction. It does not prove that the canonical D -composite system is flat, nor does it disprove a curved torsionful or relative-bimodule completion. It also does not derive the Hilbert–Palatini term or its coefficients from the fundamental UBT action.